

Learning in Firm-to-Firm Trade*

Xianglong Kong[†]
University of Chicago
Job Market Paper

Emmanuel Dhyne
National Bank of Belgium

October 20, 2025

(Updated Regularly - Latest Version [Here](#))

Abstract

We study how firms' productivity growth depends on the productivity of their suppliers and buyers and examine the aggregate implications of this form of learning. Using data on firm-to-firm transactions in Belgium, we document evidence consistent with learning from both suppliers and buyers, particularly from the most productive ones. To study the implications of this form of learning in general equilibrium, we develop a model in which firm productivity growth is shaped by the productivity of an endogenously formed network of suppliers and buyers, where firms bargain over the surpluses from trade and learning. Learning between suppliers and buyers increases bilateral trade volumes: suppliers charge lower markups when they learn more from buyers, and buyers have higher demand when they learn more from suppliers. We estimate a small open economy model with firms learning from both domestic and foreign trading partners. Quantitatively, we find that the value of learning is equivalent to 19 percent of aggregate consumption. Welfare gains from reductions in international trade participation costs are underestimated by over 7 percent without learning. Gains from trade are amplified when foreign partners are more productive through productivity catch-up and network upgrading, but are dampened when less productive foreign firms crowd out domestic learning and weaken incentives to form productive relationships.

*We are deeply indebted to Kong's thesis advisors Esteban Rossi-Hansberg, Rodrigo Adão, Chad Syverson, and Felix Tintelnot for their continuous support. We have benefited from insightful discussions with Vasco Carvalho, Juanma Castro-Vincenzi, Jonathan Dingel, Matthew Elliot, Matthias Hoelzlein, Guangbin Hong, Matthew Jackson, Benny Kleinman, Toshiaki Komatsu, Hugo Lhuillier, Feng Lin, Elisa Rubbo, Haruka Uchida, Joseph Vavra, Ruoxuan Wu, Zebang Xu, Shijian Yang, and Xiang Zhang. We thank participants at Chicago Trade Working Group, Capital Theory Seminar, Industrial Organization Working Group, Chicago Chinese Social Research Group for helpful comments. Kong is grateful to the National Bank of Belgium for data access and hospitality during his research visits. The views expressed in this paper are those of the authors and do not necessarily reflect those of the National Bank of Belgium or any other institution with which the authors are affiliated. All errors are our own.

[†]Email: xlkong@uchicago.edu. Website: <https://www.xianglong-kong.com/>.